

252-0211-00L Information Security

Semester	Spring Semester 2022
Lecturers	D. Hofheinz, S. Krstic, K. Paterson , J. L. Toro Pozo
Periodicity	yearly recurring course
Language of instruction	English



Courses

Number	Title	Hours	Lecturers
252-0211-00 V	Information Security	4 hrs Thu 14:15-16:00 HG E 7 » Fri 14:15-16:00 HG E 7 »	D. Hofheinz, S. Krstic, K. Paterson , J. L. Toro Pozo
252-0211-00 U	Information Security	3 hrs Wed 16:15-19:00 HG D 7.2 » Thu 16:15-19:00 CAB G 61 »	D. Hofheinz, S. Krstic, K. Paterson , J. L. Toro Pozo

Catalogue data

Abstract	This course provides an introduction to Information Security. The focus is on fundamental concepts and models, basic cryptography, protocols and system security, and privacy and data protection. While the emphasis is on foundations, case studies will be given that examine different realizations of these ideas in practice.
Learning objective	Master fundamental concepts in Information Security and their application to system building. (See objectives listed below for more details).
Content	1. Introduction and Motivation (OBJECTIVE: Broad conceptual overview of information security) Motivation: implications of IT on society/economy, Classical security problems, Approaches to defining security and security goals, Abstractions, assumptions, and trust, Risk management and the human factor, Course overview. 2. Foundations of Cryptography (OBJECTIVE: Understand basic cryptographic mechanisms and applications) Introduction, Basic concepts in cryptography: Overview, Types of Security, computational hardness, Abstraction of channel security properties, Symmetric encryption, Hash functions, Message authentication codes, Public-key distribution, Public-key cryptosystems, Digital signatures, Application case studies, Comparison of encryption at different layers, VPN, SSL, Digital payment systems, blind signatures, e-cash, Time stamping 3. Key Management and Public-key Infrastructures (OBJECTIVE: Understand the basic mechanisms relevant in an Internet context) Key management in distributed systems, Exact characterization of requirements, the role of trust, Public-key Certificates, Public-key Infrastructures, Digital evidence and non-repudiation, Application case studies, Kerberos, X.509, PGP. 4. Security Protocols (OBJECTIVE: Understand network-oriented security, i.e. how to employ building blocks to secure applications in (open) networks) Introduction, Requirements/properties, Establishing shared secrets, Principal and message origin

authentication, Environmental assumptions, Dolev-Yao intruder model and variants, Illustrative examples, Formal models and reasoning, Trace-based interleaving semantics, Inductive verification, or model-checking for falsification, Techniques for protocol design, Application case study 1: from Needham-Schroeder Shared-Key to Kerberos, Application case study 2: from DH to IKE. 5. Access Control and Security Policies (OBJECTIVES: Study system-oriented security, i.e., policies, models, and mechanisms) Motivation (relationship to CIA, relationship to Crypto) and examples Concepts: policies versus models versus mechanisms, DAC and MAC, Modeling formalism, Access Control Matrix Model, Roll Based Access Control, Bell-LaPadula, Harrison-Ruzzo-Ullmann, Information flow, Chinese Wall, Biba, Clark-Wilson, System mechanisms: Operating Systems, Hardware Security Features, Reference Monitors, File-system protection, Application case studies 6. Anonymity and Privacy (OBJECTIVE: examine protection goals beyond standard CIA and corresponding mechanisms) Motivation and Definitions, Privacy, policies and policy languages, mechanisms, problems, Anonymity: simple mechanisms (pseudonyms, proxies), Application case studies: mix networks and crowds. 7. Larger application case study: GSM, mobility

Performance assessment

Performance assessment information (valid until the course unit is held again)	
► Performance assessment as a semester course	
ECTS credits	8 credits
Examiners	D. Hofheinz, S. Krstic, K. Paterson, J. L. Toro Pozo
Type	session examination
Language of examination	English
Repetition	The performance assessment is only offered in the session after the course unit. Repetition only possible after re-enrolling.
Mode of examination	written 180 minutes
Written aids	Keine.

This information can be updated until the beginning of the semester; information on the examination timetable is binding.

Learning materials

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Main link Information

Only public learning materials are listed.

Groups

No information on groups available.

Restrictions

There are no additional restrictions for the registration.

Offered in

Programme	Section	Type	
Data Science Master	Core Electives	W	
Computer Science Bachelor	Major: Theoretical Computer Science	O	
Computational Science and Engineering Bachelor	Electives	W	

